

Course „Theoretical Ecotoxicology: Concepts and applications“

Session 15: Introduction to spatial analysis

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[Satellite Embeddings](#)

About me

B.Sc. in Environmental Management at Justus-Liebig-University Giessen

-> Hydrodechlorination of mine water-specific PCBs using palladium nanoparticles

M.Sc. in Environmental Sciences at Justus-Liebig-University Giessen

-> Statistical analyses of antibiotics in Hessian surface waters (2012-2024)

Currently doing PhD in the in-silicio part of the Ecotox department

-> Large scale relationships between anthropogenic chemicals and risks in a One Health context



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[University Giessen](#)

Learning objectives

“Why spatial analysis”: interesting (ecotox+geospatial)-related case studies

„What is geographic data“: vectors, rasters and coordinates

„How do we get geographic data“: different geographic data sources

“What can we do with geographic data”: spatial, geometric and vector-raster operations

“Statistical learning with geographic data”: patterns, predictions and causes



© Cover art for „[Spatial Data Science with applications in R](#)“
by [Allison Horst](#)

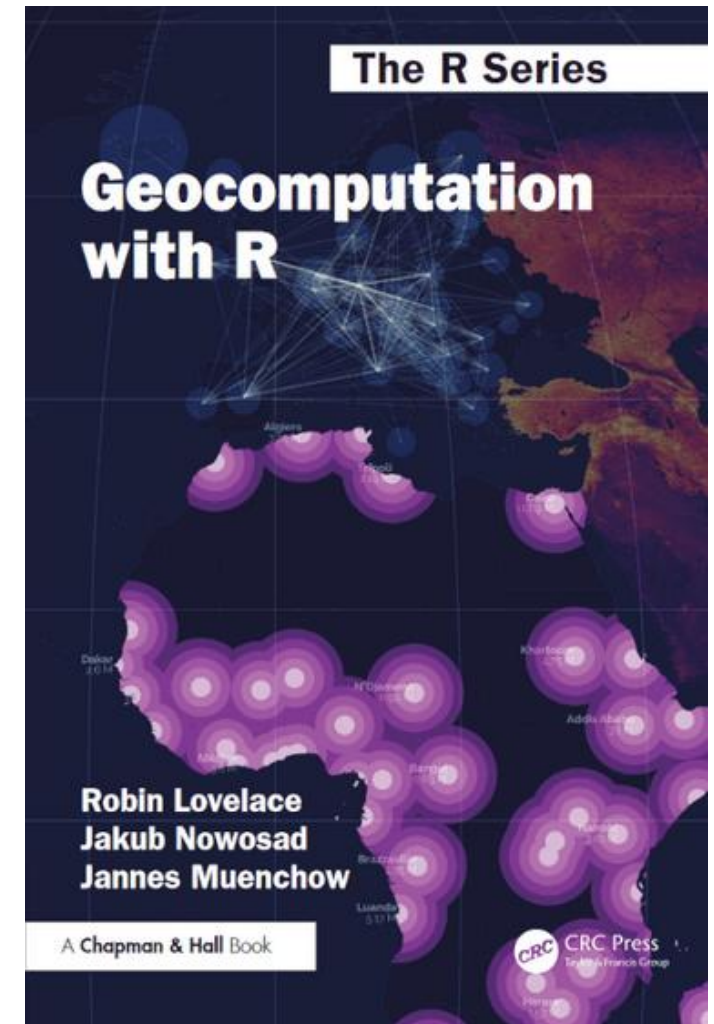
Geocomputation with R

Geocomputation:

Research that uses geographic data to solve problems, with a focus on reproducibility, flexibility and tool development.

Why with R?

R has the ability to seamlessly switch between geographic and non-geographic data to process, model and visualize while enabling reproducibility and the usage of a multitude of Geographic Information System (GIS) processes and plug-ins.

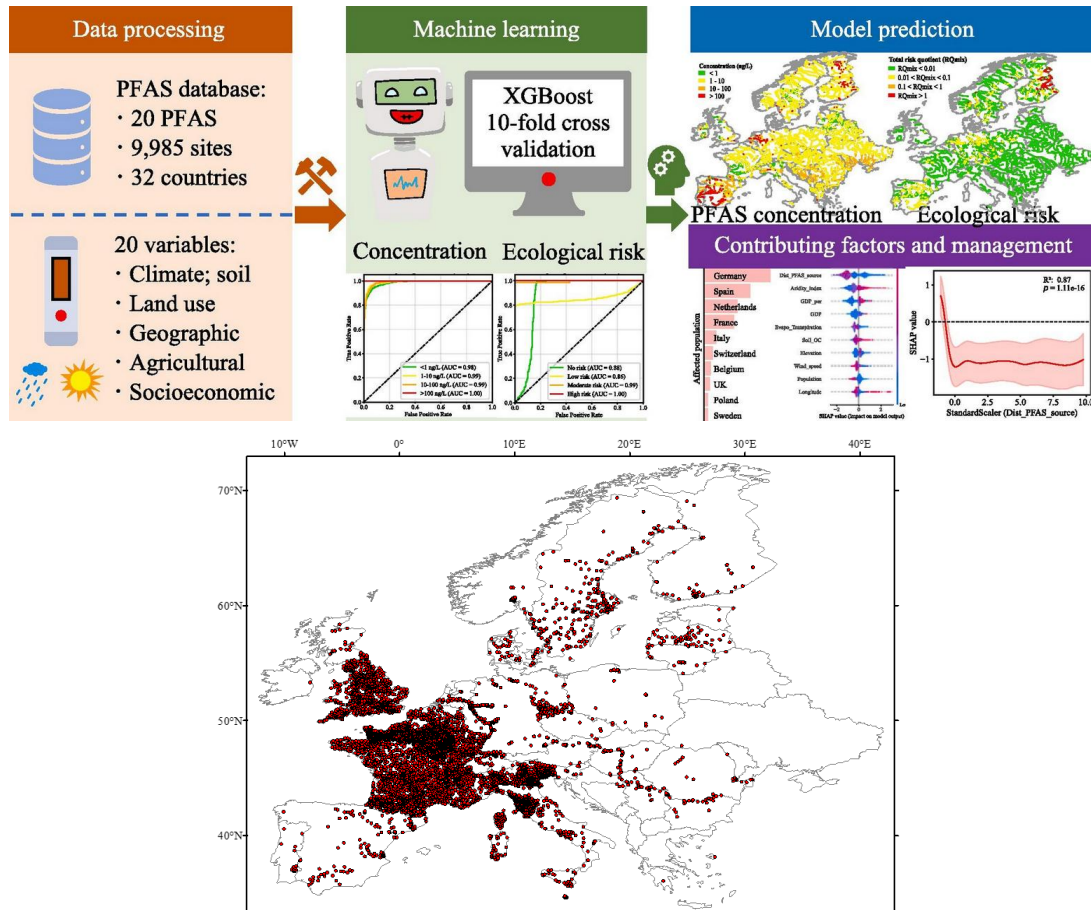


<https://r.geocompx.org/>

Why spatial analysis? (1/3)

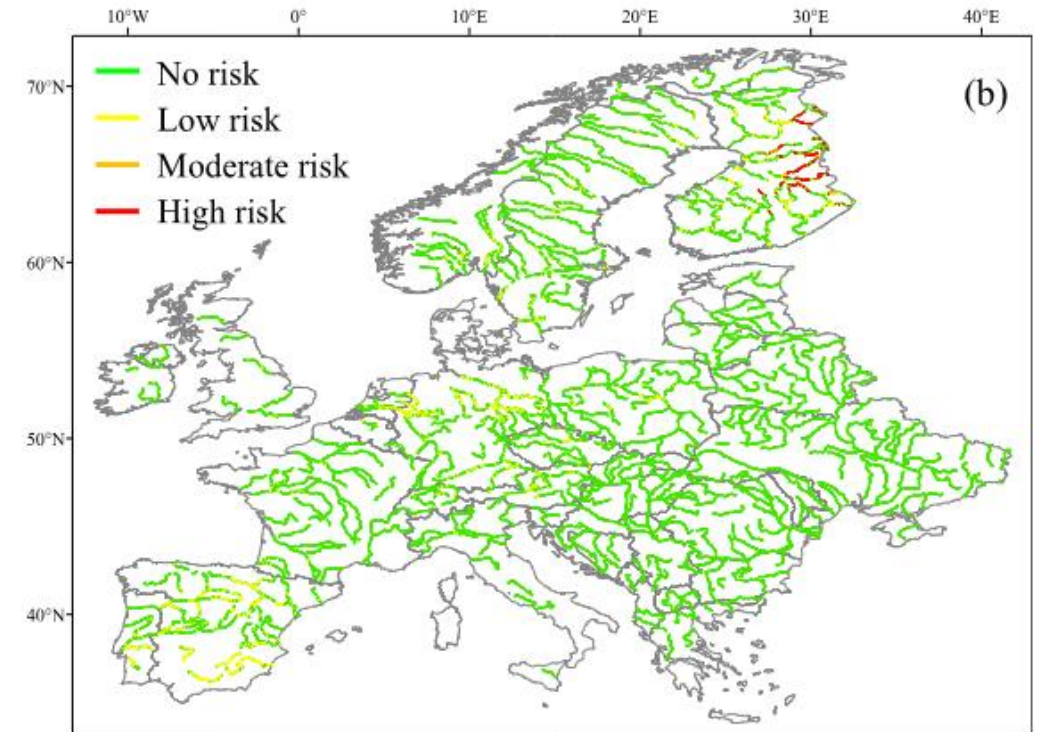
Unveiling PFAS hazard in European surface waters using an interpretable machine-learning model

Li Zhao^{a,b}, Jian Chen^{c,*}, Jiaqi Wen^d, Yangjie Li^{a,*}, Yingjie Zhang^c, Qunyue Wu^a, Gang Yu^e



What is the spatio-temporal distribution of emerging contaminants?

Using the power of **ML** to **predict** concentrations and ecological risks from known to unknown locations (retrospective)!



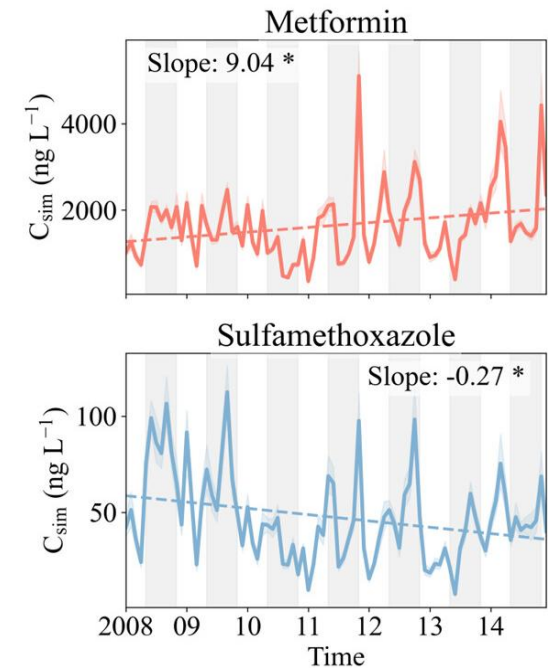
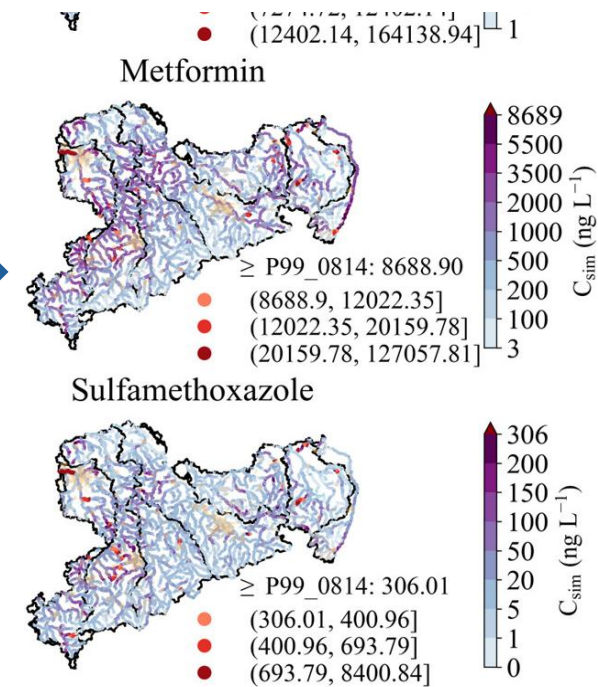
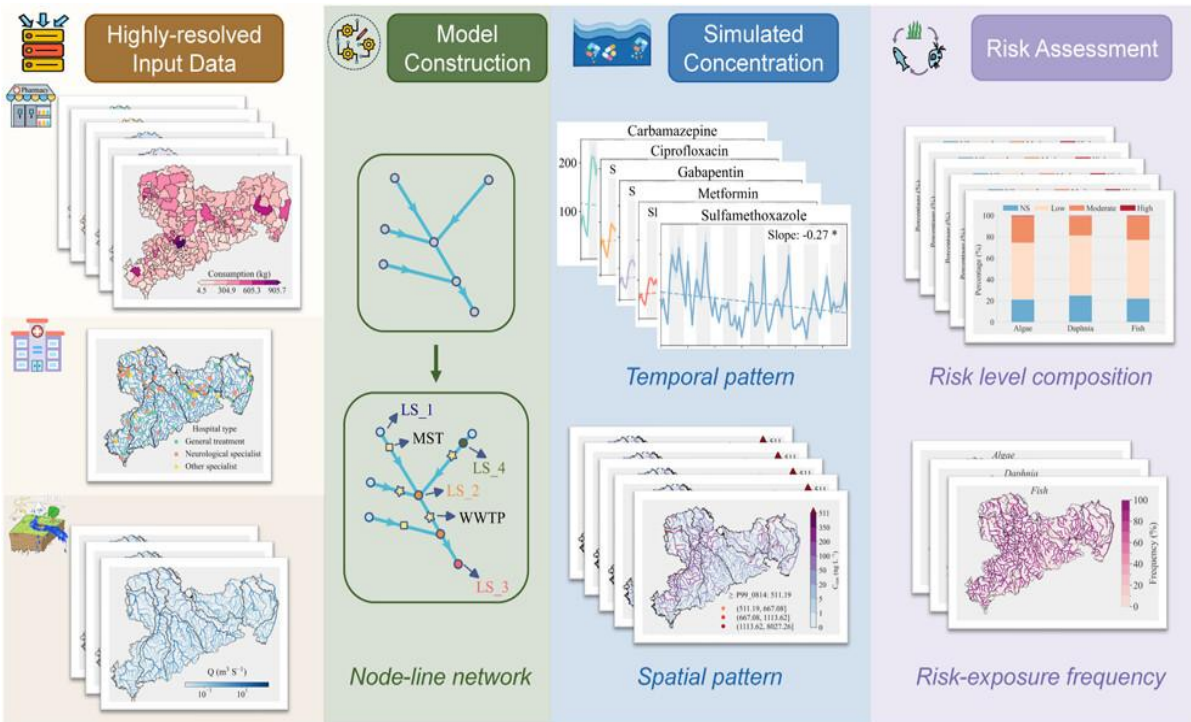
Why spatial analysis? (2/3)

Spatially and Seasonally Resolved Predictions Reveal Widespread Ecotoxicological Risk from Pharmaceutical Mixtures in German (Saxon) Rivers

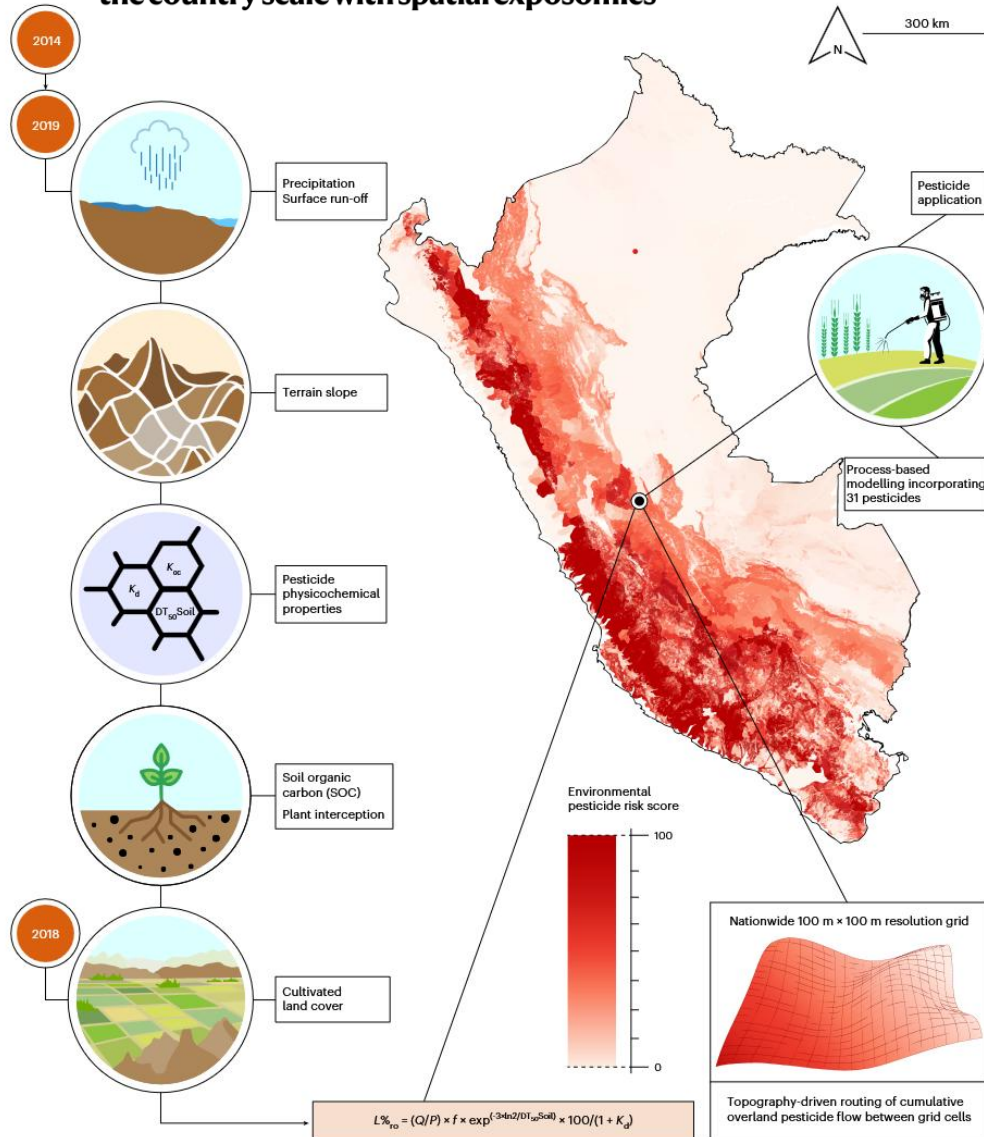
Shixue Wu,* Björn Helm, Geovanni Teran-Velasquez, Peter Krebs, and Rohini Kumar*

What is the spatio-temporal distribution of emerging contaminants?

Using the power of **contaminant fate models** to **predict** concentrations and ecological risks from process-based knowledge (retro- & prospective)!

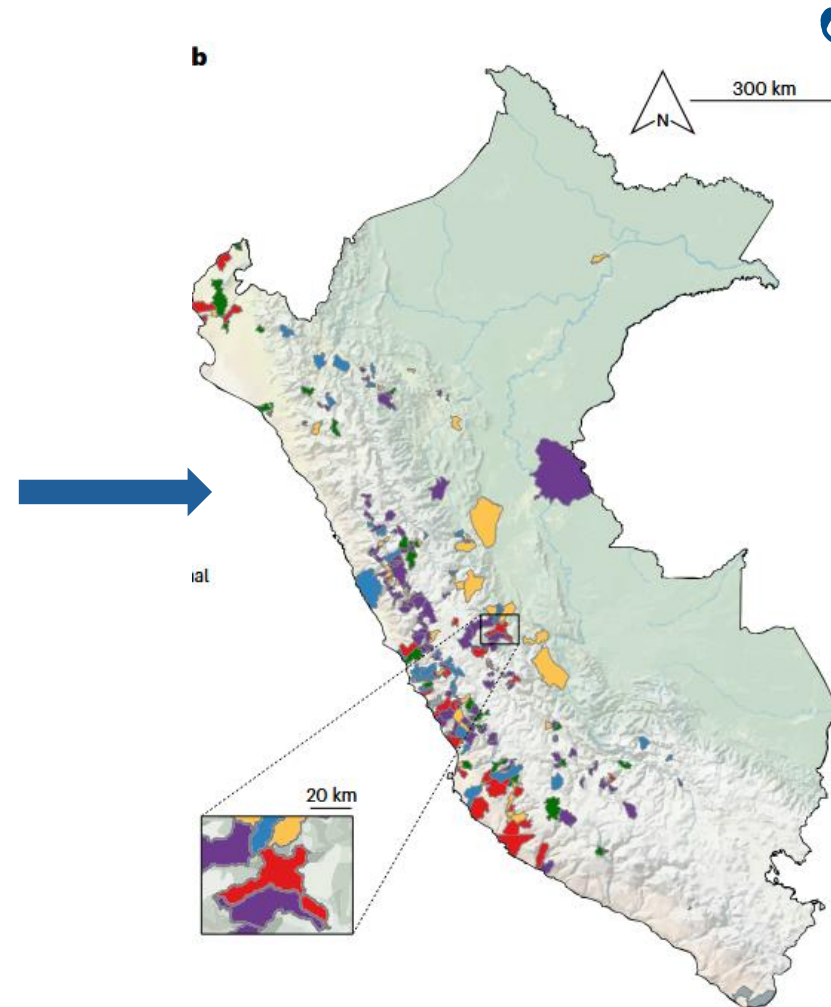


Mapping pesticide mixtures to cancer risk at the country scale with spatial exposomics



Why spatial analysis? (3/3)

What is the spatio-temporal distribution of emerging contaminants?



Linking exposure & health data through **integrative frameworks** to understand risks in an One Health context!

What is geographic data? (1/4)

In a nutshell, geographic data has a spatial component attached to its attribute table!

There are, however, two different fundamental **geographic data models** behind geographic data: rasters and vectors.



<https://r-spatial.github.io/sf/>



<https://r-spatial.github.io/terra/index.html>



<https://r-spatial.github.io/qgisprocess/>

What is geographic data? (2/4)

Raster

- Represents the world with continuous grid of cells (= pixels);
- Better for continuous phenomena such as elevation, temperature, ... ;
- TIFF, NetCDF, Cloud Optimized GeoTIFF, ... ;
- terra-package,

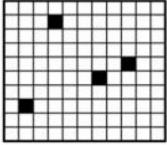
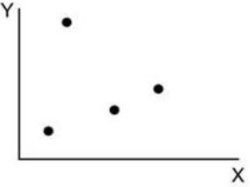
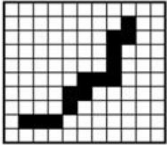
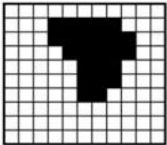
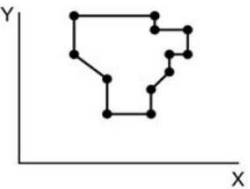
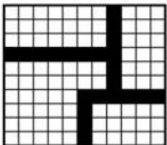
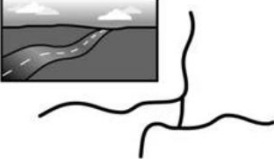
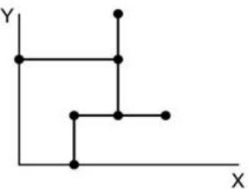
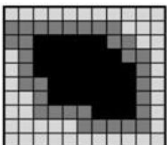
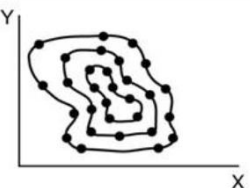
```
my_rast
#> class      : SpatRaster
#> size       : 457, 465, 1  (nrow, ncol, nlyr)
#> resolution : 0.000833, 0.000833  (x, y)
#> extent     : -113, -113, 37.1, 37.5  (xmin, xmax, ymin, ymax)
#> coord. ref.: lon/lat WGS 84 (EPSG:4326)
#> source     : srtm.tif
#> name       : srtm
#> min value  : 1024
#> max value  : 2892
```

Vector

- Represents the world with points, lines and polygons;
- Better for discrete, well-defined borders such as WWTP, human settlements, ... ;
- ESRI Shapefile, Geopackage, ... ;
- sf-package,

```
world_mini = world[1:2, 1:3]
world_mini
#> Simple feature collection with 2 features and 3 fields
#> Geometry type: MULTIPOLYGON
#> Dimension: XY
#> Bounding box: xmin: -180 ymin: -18.3 xmax: 180 ymax: -0.95
#> Geodetic CRS: WGS 84
#> # A data frame: 2 x 4
#>   iso_a2 name_long continent geom
#> * <chr> <chr> <chr> <MULTIPOLYGON [°]>
#> 1 FJ Fiji Oceania (((-180 -16.6, -180 -16.5, -180 -16, -180 -16.1, -180 -16.2, -180 -16.3, -180 -16.4, -180 -16.5, -180 -16.6, -180 -16.7, -180 -16.8, -180 -16.9, -180 -17, -180 -17.1, -180 -17.2, -180 -17.3, -180 -17.4, -180 -17.5, -180 -17.6, -180 -17.7, -180 -17.8, -180 -17.9, -180 -18, -180 -18.1, -180 -18.2, -180 -18.3, -179.9 -18.3, -179.9 -18.2, -179.9 -18.1, -179.9 -18, -179.9 -17.9, -179.9 -17.8, -179.9 -17.7, -179.9 -17.6, -179.9 -17.5, -179.9 -17.4, -179.9 -17.3, -179.9 -17.2, -179.9 -17.1, -179.9 -17, -179.9 -16.9, -179.9 -16.8, -179.9 -16.7, -179.9 -16.6, -179.9 -16.5, -179.9 -16.4, -179.9 -16.3, -179.9 -16.2, -179.9 -16.1, -179.9 -16, -179.9 -15.9, -179.9 -15.8, -179.9 -15.7, -179.9 -15.6, -179.9 -15.5, -179.9 -15.4, -179.9 -15.3, -179.9 -15.2, -179.9 -15.1, -179.9 -15, -179.9 -14.9, -179.9 -14.8, -179.9 -14.7, -179.9 -14.6, -179.9 -14.5, -179.9 -14.4, -179.9 -14.3, -179.9 -14.2, -179.9 -14.1, -179.9 -14, -179.9 -13.9, -179.9 -13.8, -179.9 -13.7, -179.9 -13.6, -179.9 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```

What is geographic data? (3/4)

The raster view of the world	Happy Valley spatial entities	The vector view of the world
	 x x Points: hotels	
	 Lines: ski lifts	
	 Areas: forest	
	 Network: roads	
	 Surface: elevation	

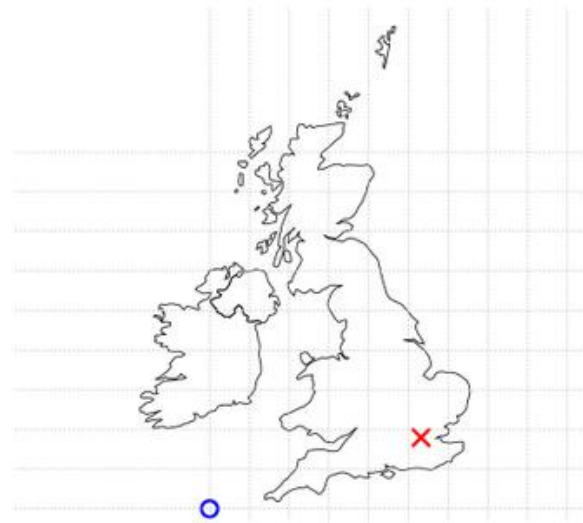
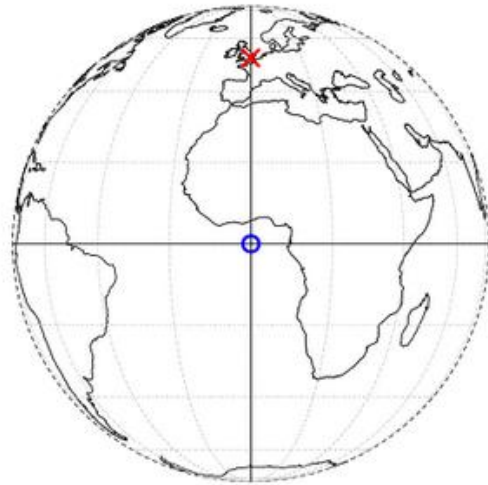
"An Introduction to Geographical Information Systems" by Ian Heywood, Sarah Cornelius, and Steve Carver

What is geographic data? (4/4)

Coordinate Reference System (CRS): defines how the spatial elements of the data relate to the surface of Earth. There are two different CRS types: geographic (lat/lon, with units in degrees) or projected (typically with units of meters from a datum).

Geographic CRS: WGS84
(EPSG: 4326)

→ Most used in the world



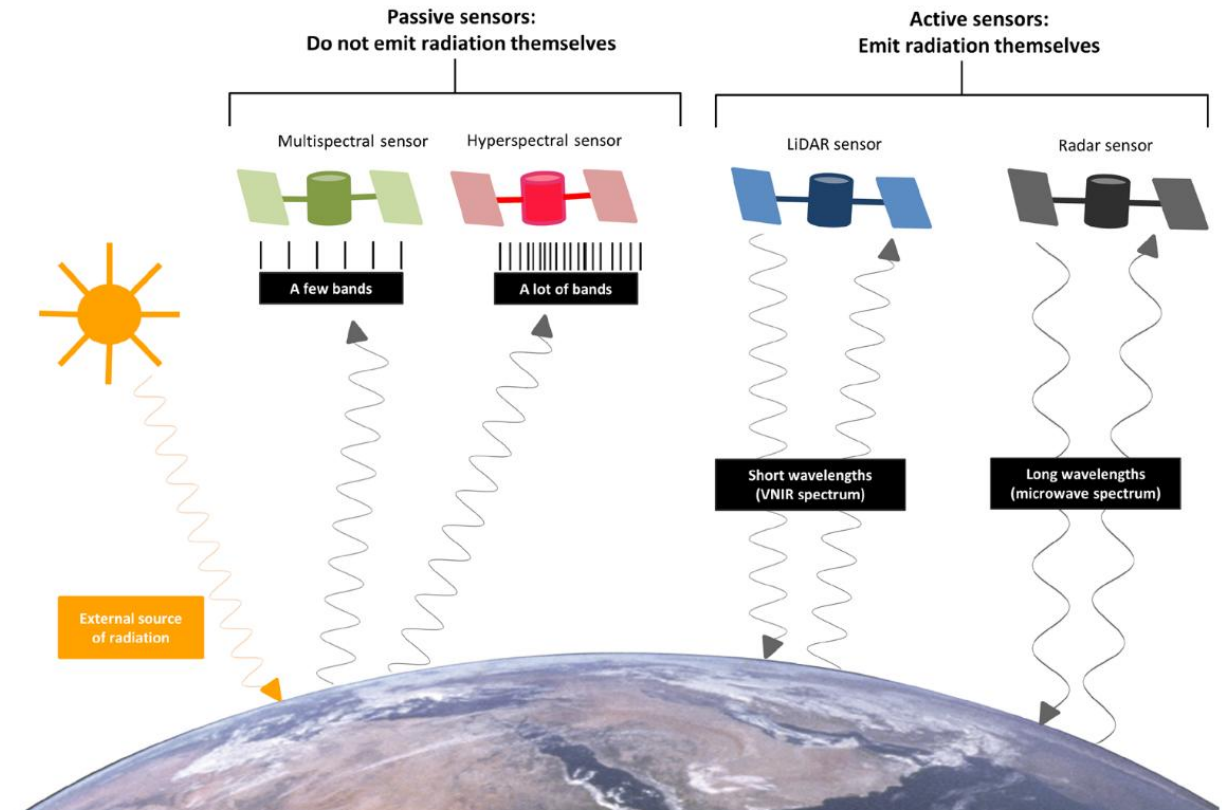
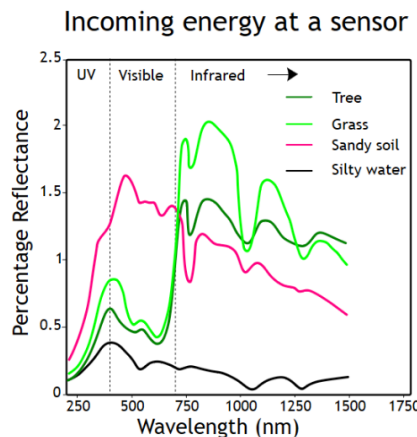
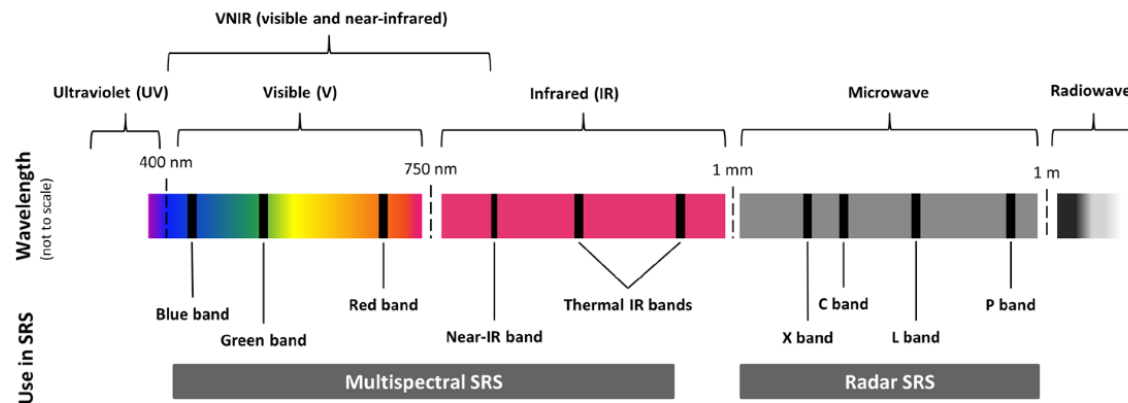
Projected CRS: ETRS89 /
UTM zone 32N
(EPSG: 25832)

→ Most used in Germany

FIGURE 2.1: Vector (point) data in which the location of London (red X) is represented with reference to an origin (blue circle). The left plot represents a geographic CRS with an origin at 0° longitude and latitude. The right plot represents a projected CRS with an origin located in the sea west of the South West Peninsula.

How do we get geographic data? (1/4)

A big part of geographic data bases on satellites and **remote sensing** (RS). **RS** is the process of identifying, observing, and measuring an object without coming into contact with it. **Satellite** RS measures electromagnetic radiation reflected from objects, which is influenced by the objects size, shape, orientation, temperature, and chemical composition.



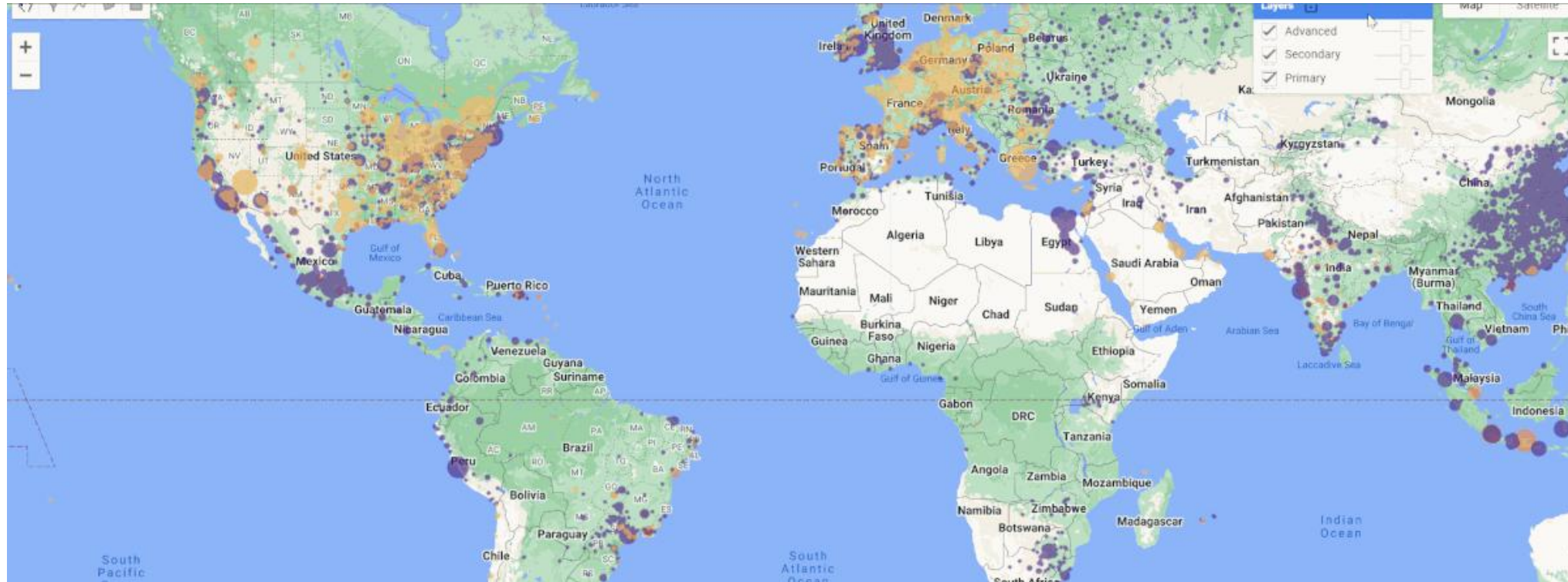
How do we get geographic data? (2/4)

Besides RS, you can get geographic data based on **in-situ measurements** (e.g. monitoring stations), **citizen science** (e.g. iNaturalist or HydroCrowd), **administrative data** (e.g. EU-NUTS or GDP), **scientifically produced open data** and the list goes on.

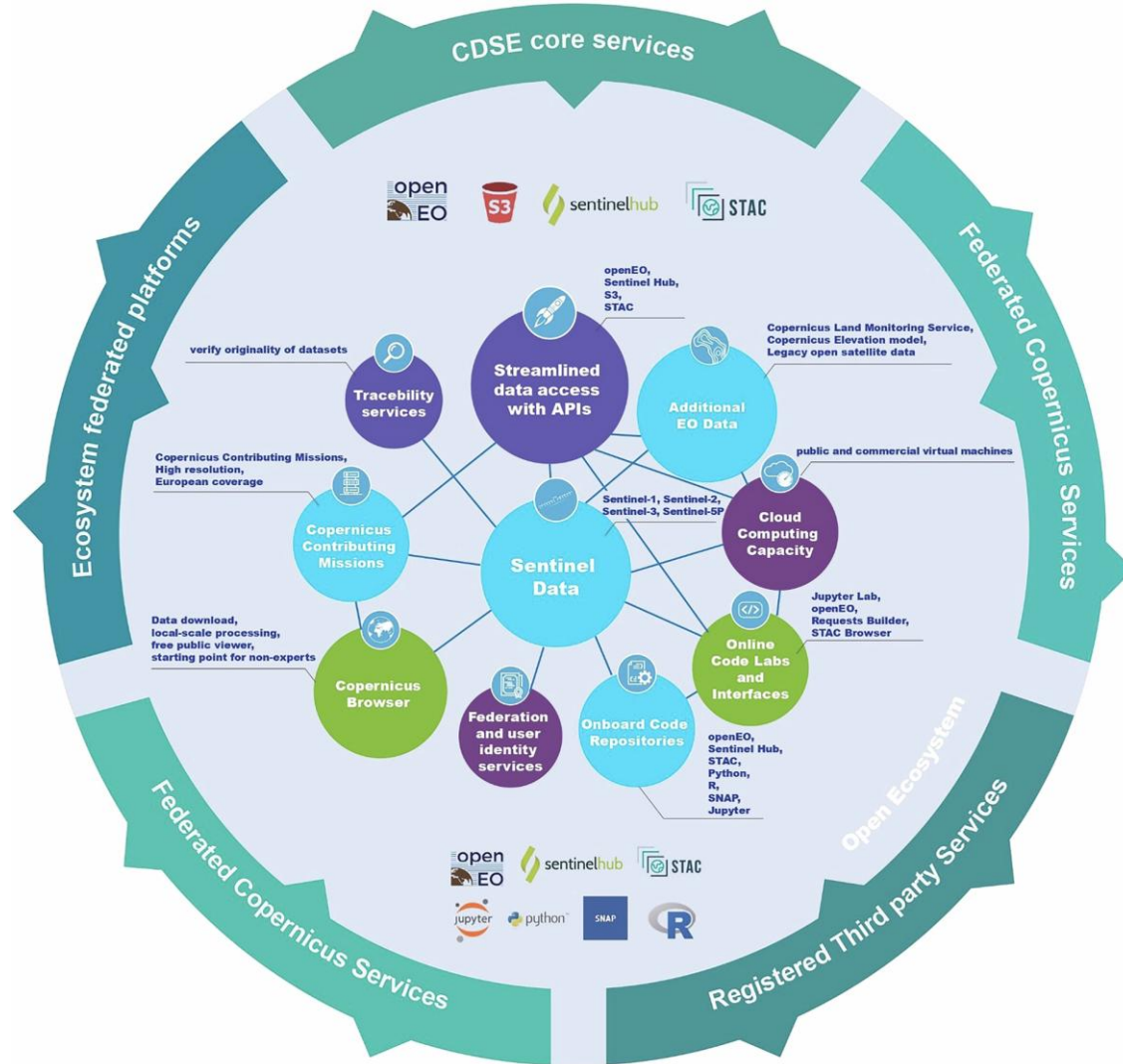
Remote sensing	In-situ measurements	Scientific open data
SpatioTemporal Asset Catalogs (STAC)	Curated NORMAN (EU)	WorldClim (bio-climatic parameters)
Google Earth Engine (GEE)	Waterbase (EU)	EarthENV (topographic parameters)
Awesome GEE Catalog (Awesome-GEE)	WATCH (DE) - to be released	HydroWASTE (spatial & technical WWTP data)
Copernicus Data Space Ecosystem (CDSE)	Forever Pollution Project (EU)	Google Earth Engine Embeddings (remote sensing & climate embedding)
...	...	...

How do we get geographic data? (3/4)

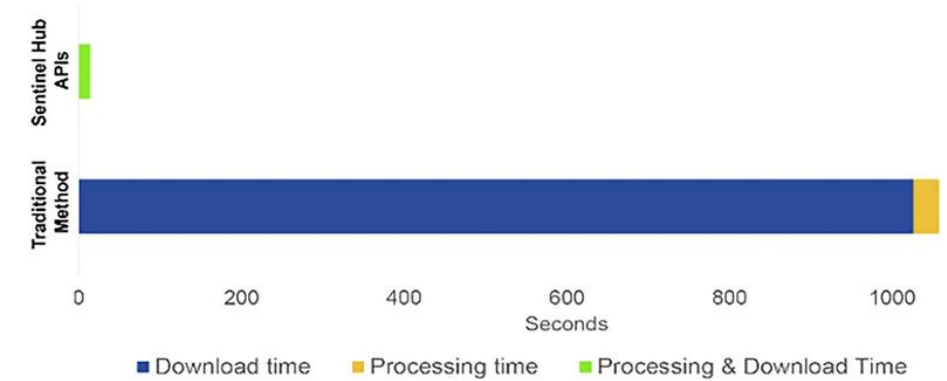
Example for geographical data on awesome-google-earth-engine-catalog: [HydroWASTE](#)



How do we get geographic data? (4/4)



Data Processing & Download Time



What can we do with geographic data? (1/3)

Spatial data operations means manipulating the geographic data based on their non-geographic attributes and spatial relations (**joining, intersecting, overlapping, map algebra, et cetera**). Examples for such operations on vectors (left) and rasters (right).

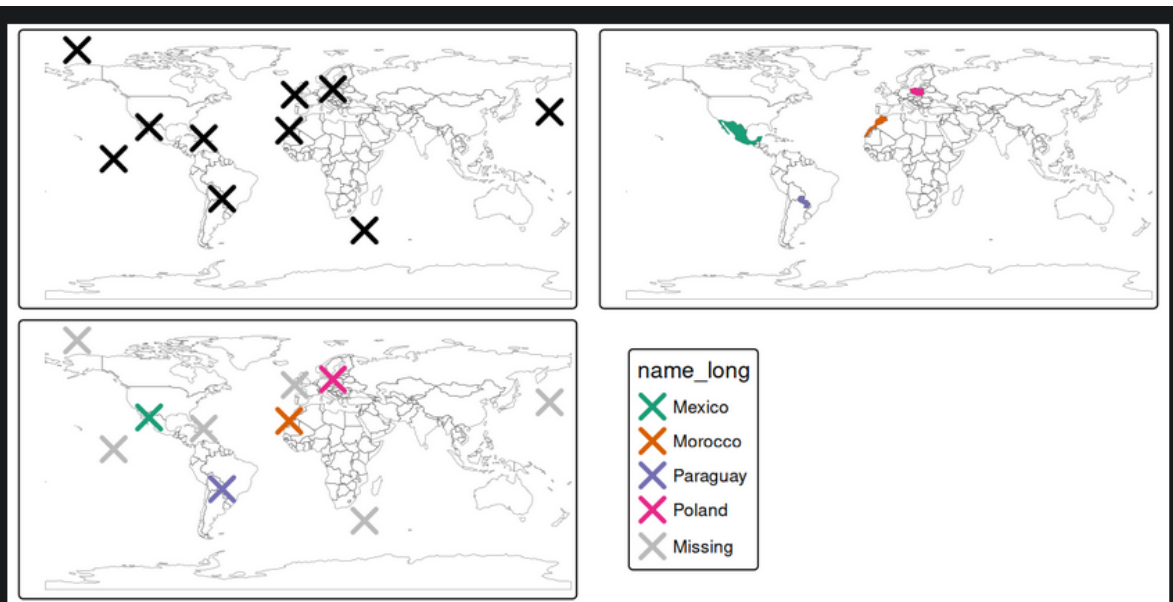


FIGURE 4.6: Illustration of a spatial join. A new attribute variable is added to random points (top left) from source world object (top right) resulting in the data represented in the final panel.

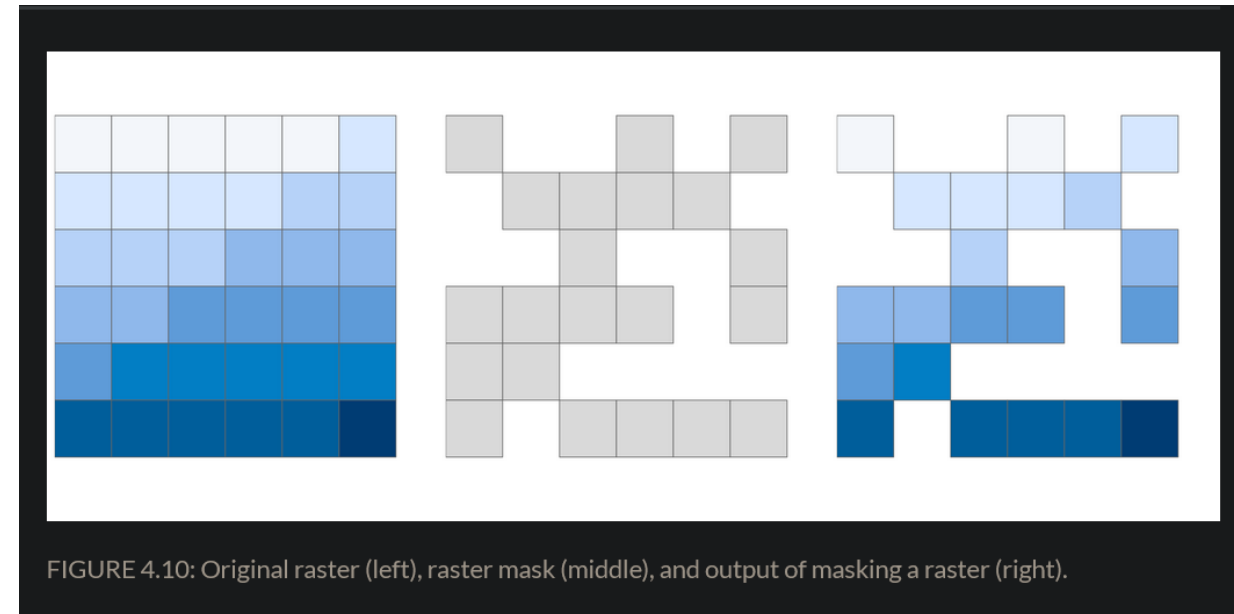
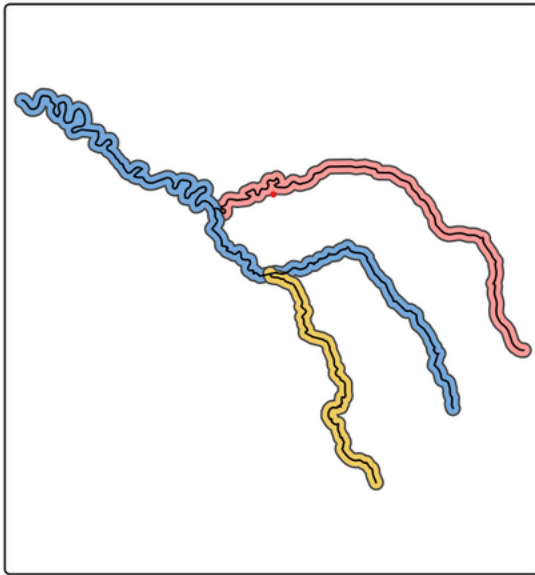


FIGURE 4.10: Original raster (left), raster mask (middle), and output of masking a raster (right).

What can we do with geographic data? (2/3)

Geometry operations means manipulating the geographic data based their geographic element (**buffering, resampling, simplifying, converting, et cetera**). Examples for such operations on vectors (left) and rasters (right).

5 km buffer



50 km buffer

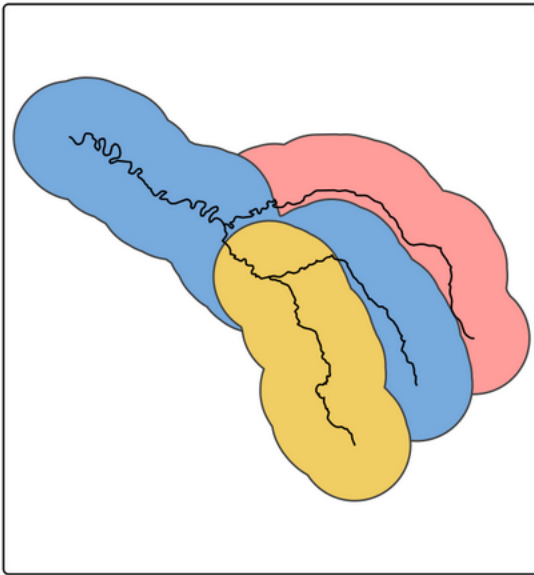
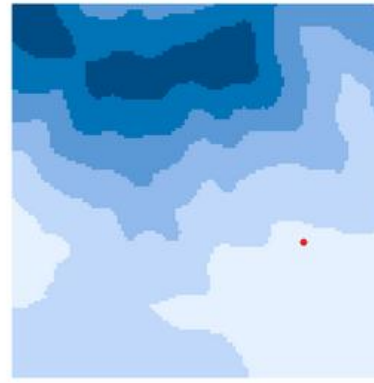
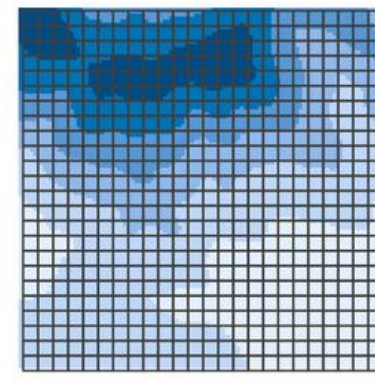


FIGURE 5.4: Buffers around the Seine dataset of 5 km (left) and 50 km (right). Note the colors, which reflect the fact that one buffer is created per geometry feature.

Original raster



Target raster



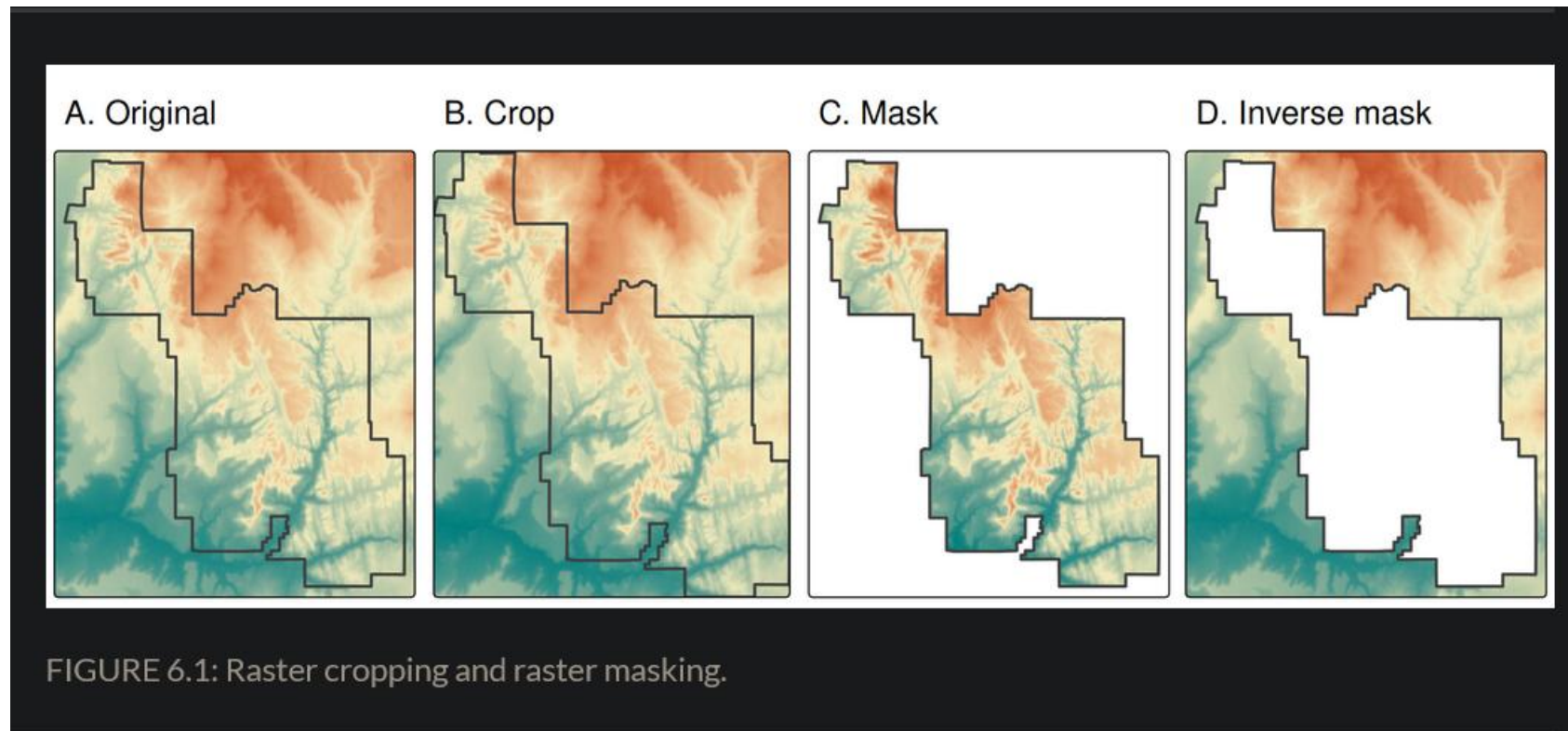
Resampled raster



FIGURE 5.17: Resampling of an original (input) raster into a target raster with custom resolution and origin.

What can we do with geographic data? (3/3)

Many geographic data projects involve integrating data from different sources, e.g. remote sensing images (rasters) and administrative boundaries (vectors). In such cases, we make use of **raster-vector operations (extracting, cropping, converting, et cetera)**. Examples for such operations down beneath.



How it feels to have a conversation when everyone knows what GIS stands for

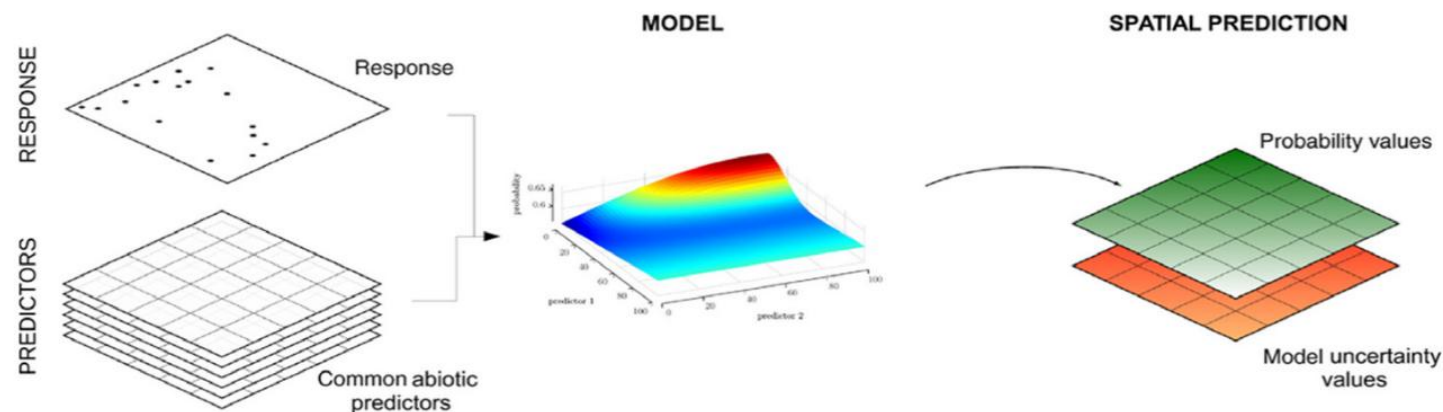


<https://i.redd.it/92rs7g7ls3gb1.jpg>

Questions?

Statistical learning with geographic data (1/4)

Statistical learning is concerned with the use of statistical and computational models for **identifying patterns in data** and **predicting from these patterns**.



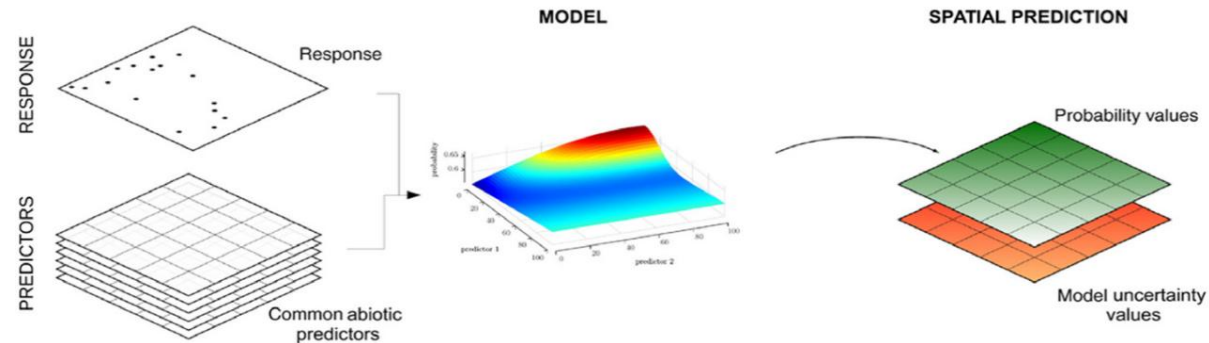
Response: Biological data (species-plot data).

Predictors: Remote sensing (NDVI) & topographic (elevation, catchment slope and area) variables.

Goal: Predict floristic gradient spatially.

Statistical learning with geographic data (2/4)

LIVE DEMONSTRATION



THEORETICAL USAGES



Response: Biological data (species-plot data).

Predictors: Remote sensing (NDVI) & topographic (elevation, catchment slope and area) variables.

Goal: Predict floristic gradient spatially.

Response: Exposure, consumption, epidemiology data,

Predictors: Climatic, topographic, biogeographic, socio-economic, remote sensing, technical, biological variables,

Goal: Predict ecological/health risks from emerging contaminants!

Statistical learning with geographic data (3/4)

On the one hand,

one has to question themselves regarding the **quality of the prediction** (area of applicability, consideration of spatial autocorrelation, et cetera).

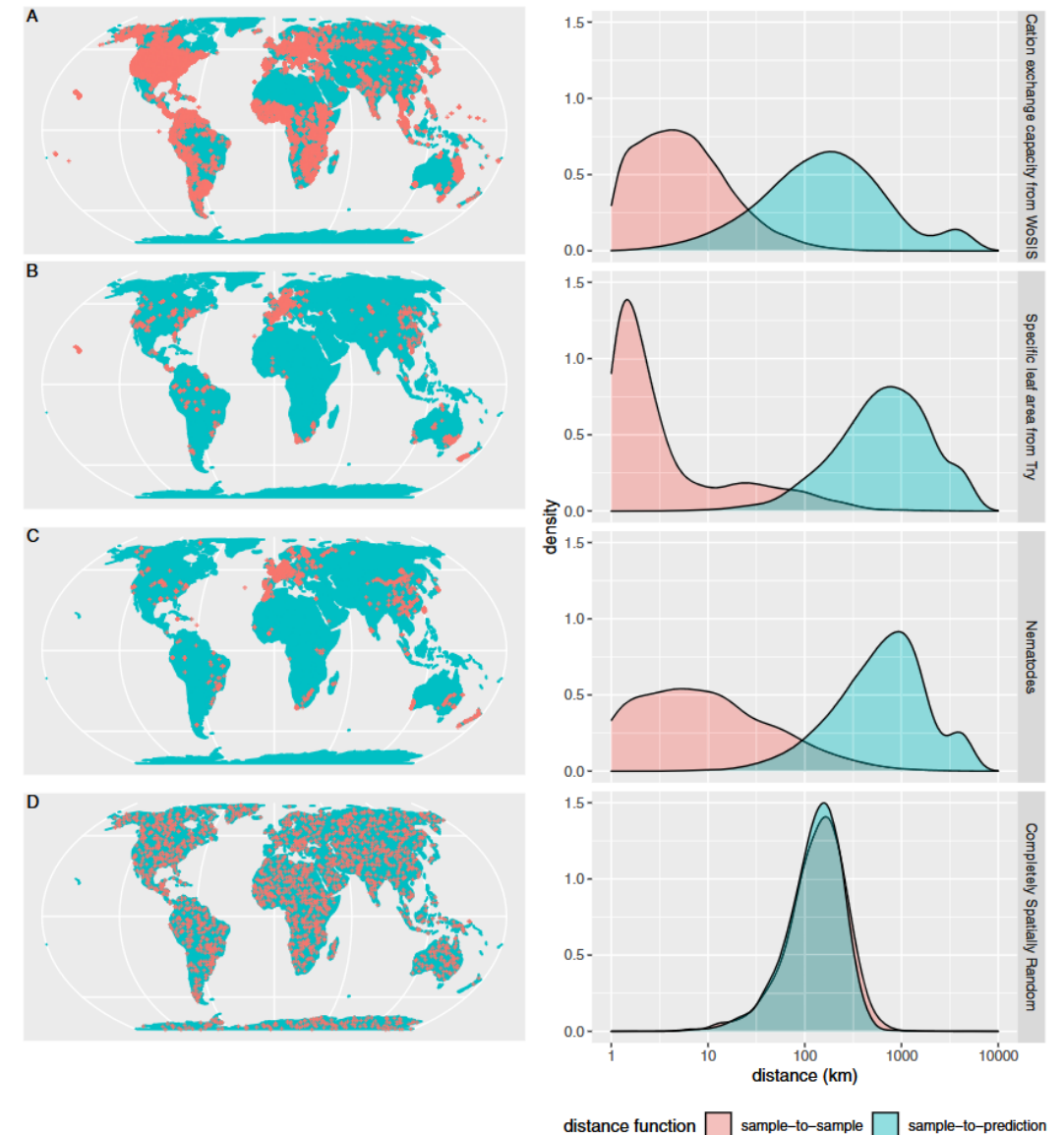
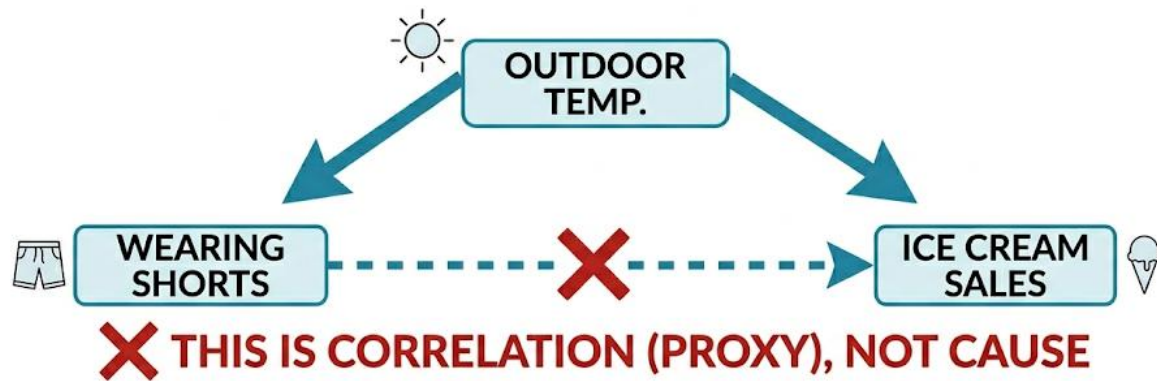


Fig. 1 Spatial distance distributions in global mapping studies. Spatial distribution (left; equal Earth projection) and distribution of nearest neighbor distances (right; sample-to-sample distance in pink, prediction-location-to-sample distance in blue) for three different publicly available datasets: cation exchange capacity in the soil from the WoSIS database²³ as used for global soil mapping³ (A), specific leaf area from the TRY database²⁴ as used for the global mapping in Moreno-Martinez et al. (2018)²⁵ (B), and the nematodes dataset compiled by Van den Hoogen et al. (2019)² (C). For comparison, the fourth dataset is a simulated completely spatially random sample of the same size as the nematode dataset (D). Distance distributions were calculated and visualized using the R package "CAST"²⁶.

Statistical learning with geographic data (4/4)

CORRELATION vs. CAUSATION ICE CREAM & SHORTS



✓ OK FOR
PREDICTING SALES

STOP BAD FOR
CHANGING SALES

On the other hand,

one has to think about the **meaning of the prediction** (research question, causal inference, et cetera).

© Created with the assistance of AI (Gemini), based on „The Effect“ by Huntington-Klein (2021)

“Why spatial analysis”

case studies showing the power of different geospatial models to predict the spatio-temporal distribution and risks of contaminants

„What is geographic data“

importance of vectors, rasters and coordinate reference systems

„How do we get geographic data“

satellite remote sensing and different geographic data sources, including cloud computing

“What can we do with geographic data”

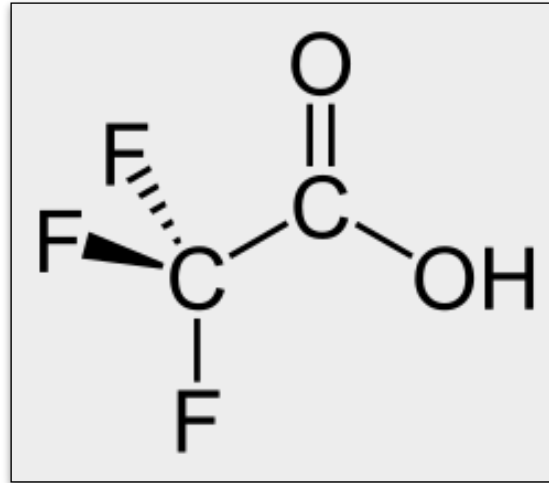
introductory informations about spatial, geometric and vector-raster operations

“Statistical learning with geographic data”

patterns, predictions and causes

Widespread contamination of German shallow groundwaters through Trifluoroacetic acid

Belongs to the subclass of per- and polyfluoroalkyl substances (PFAS) known as **ultrashort-chain perfluoroalkyl acids (PFAAs)**

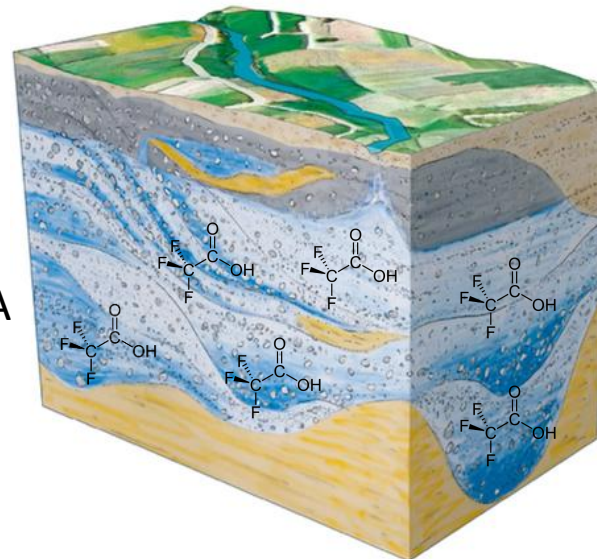


Most abundant PFAS (detection rate & concentration) in the environment

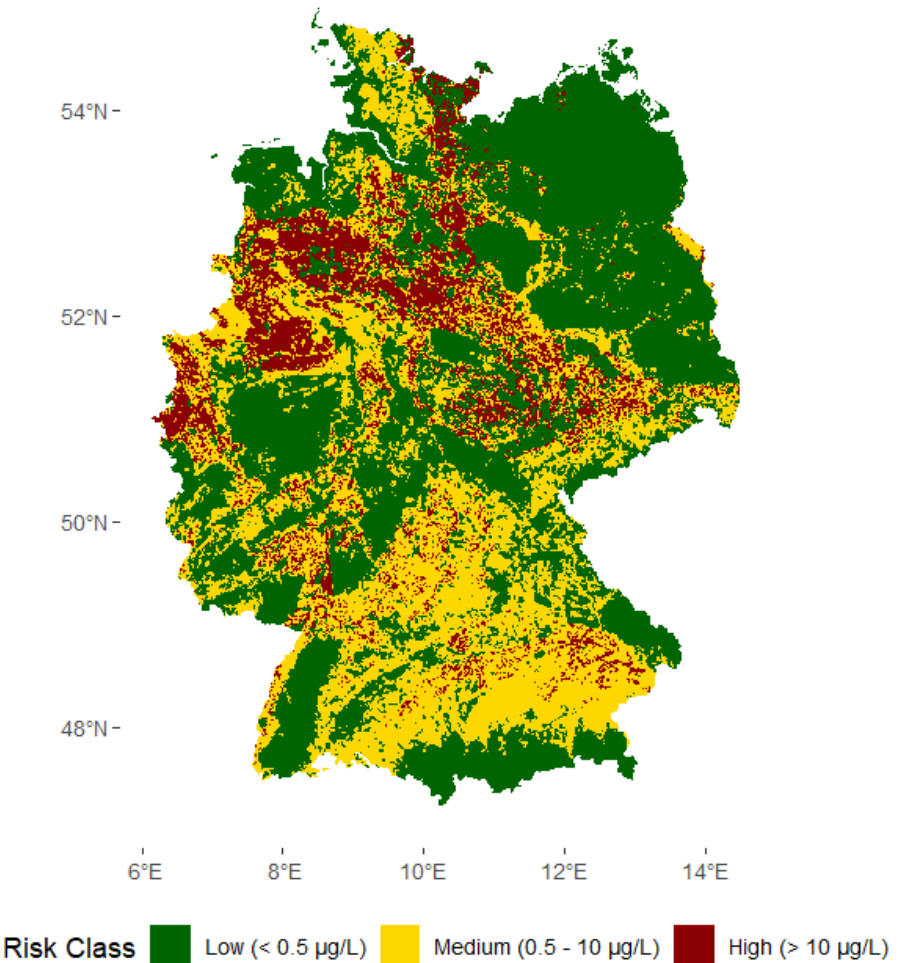
Groundwater can be a long-term sink for TFA

→ Persistent, Mobile, and Toxic (**PMT**)

→ very Persistent and very Mobile (**vPvM**)



TFA risk in shallow (< 100 m) German groundwaters





© Google Earth Engine
Satellite Embeddings

Thank you for your attention!

PhD topic: “Large scale relationships between anthropogenic chemicals and risks in a One Health context”

Contact: <https://www.uni-due.de/biologie/oekotoxikologie/bobric-silviu.php>